

## ECS401: Unit 8: Writing programs Progress Test: Arrays of Records Solution / Feedback

Here are the two missing methods

```
// Input and store the details of 50 cars
//
public static void inputCars(Car [] c)
{
    for(int i=0; i<c.length; i++)
    {
        String reg = input("Registration?");
        String make = input("Make?");
        Car nextcar = createCar(reg, make);
        c[i] = nextcar;
    }
    return;
}

// create a car record with registration and year
//
public static Car createCar (String reg, String mk)
{
    Car c = new Car();
    c.registration = reg;
    c.make = mk;
    return c;
}
```

The first method, `inputCars`, is given an array of cars as argument which it calls `c` locally. It then runs a for loop that inputs a car 50 times and stores it in the array, `c`, at the next index position. The body of the loop just inputs the two Strings for the make and registration. It uses the `createCar` method to store them in a newly created record. That record returned from `createCar`, is then stored first in `nextCar` and then copied into the array in position `i` (the next index position). Note this passes the array by reference so on return the original array passed (in variable `cars` in main) is filled.

The `createCar` method first uses `new` to create an empty record of the right type and stores it in variable `c`. It then assigns the given values to the appropriate fields of `c` and returns the new `Car` record.